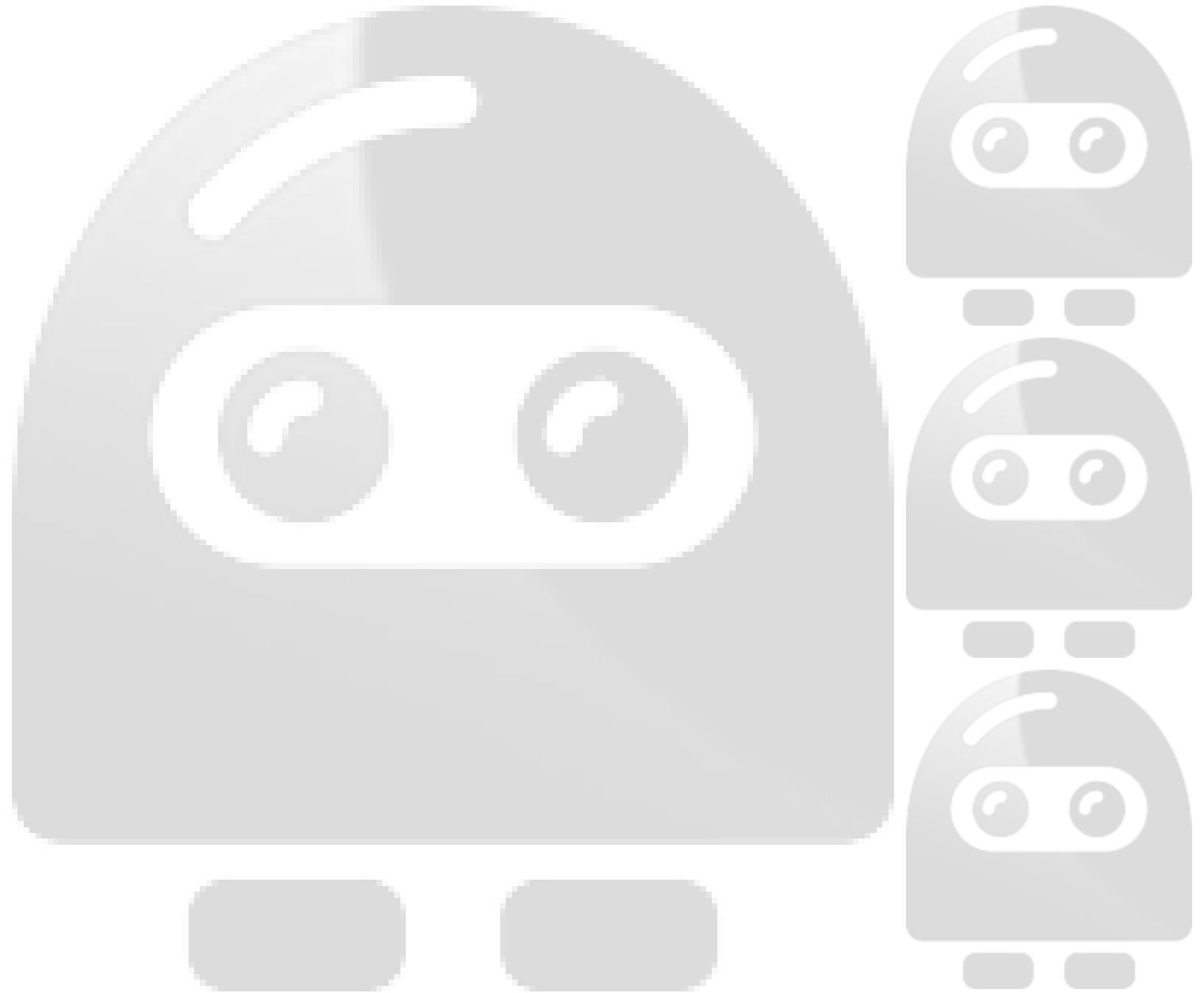


BPM
CSE346
RPA



Dr. Islam El-Maddah

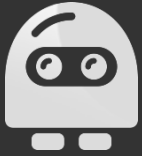
DIGITAL BUSINESS AUTOMATION

What is RPA?

Robotic Process Automation is the innovative use of software to perform repetitive rules-based knowledge work across an organization as a substitute for, or **aide to, human workers**.

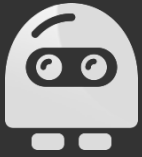
“Virtual Workers” replicate the specific actions a human would take while working with IT systems, the decisions they make, and the logical processes they follow, while interacting between different systems and applications.





45% of all work

activities can be automated



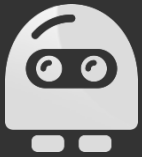
\$2 trillion dollars

could be saved in workforce spending



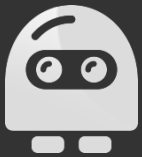
6-9 Months

on average ROI for an RPA project



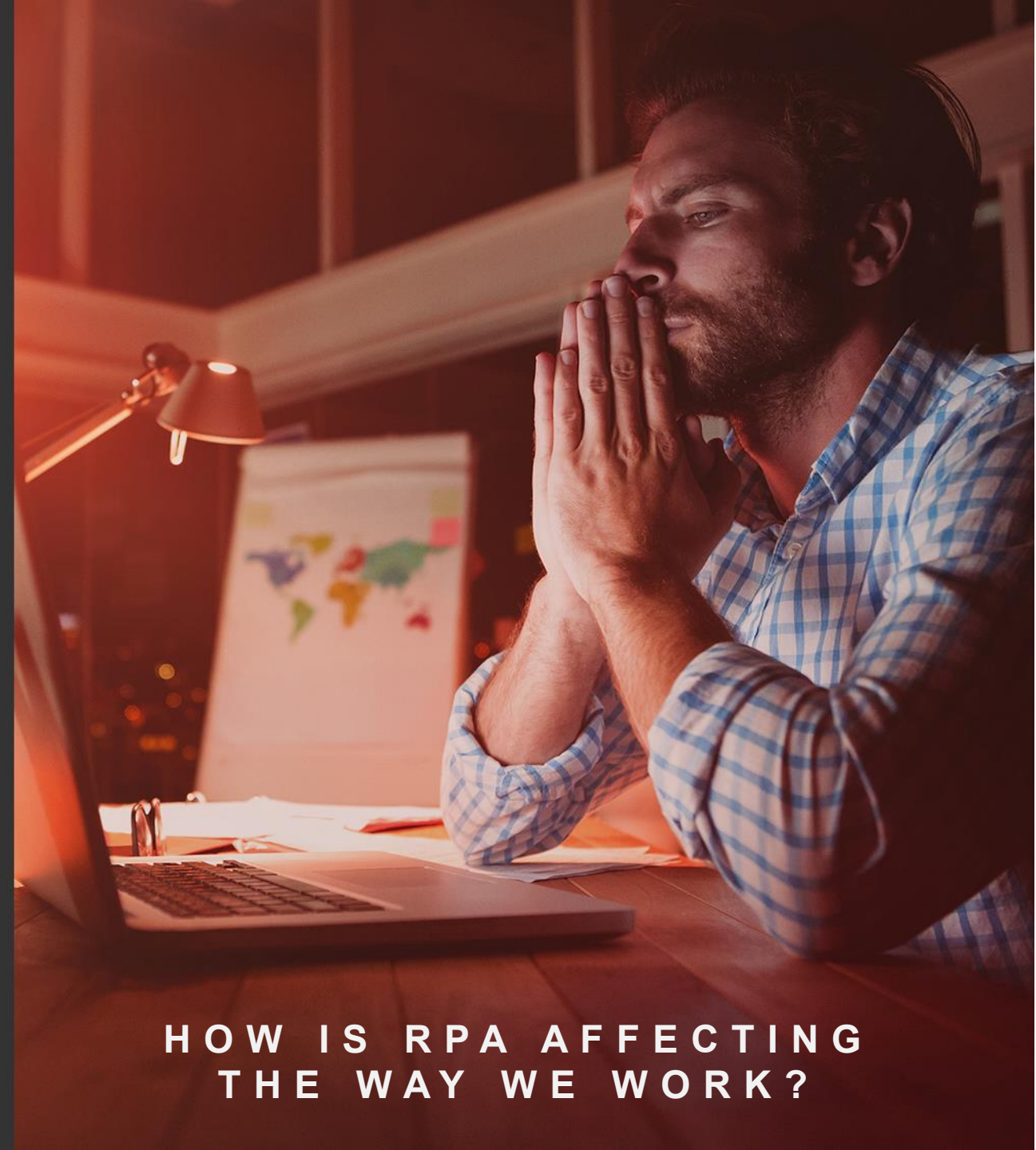
94% of CEOs

that have incorporated RPA believe
that it has increased the productivity



from 2025

25% of jobs are robotized



**HOW IS RPA AFFECTING
THE WAY WE WORK?**



RPA

BUSINESS BENEFITS



Government and regulatory compliance



Virtual workforce 24x7x365



Productivity increase



Reduction of human errors



Flexibility



Cost reduction



Scalability



RAPID IMPLEMENTATION AND ROI
REDUCES THE NUMBER OF FTES IN A PROCESS

RPA TYPICAL BUSINESS PROCESSES

RPA can learn from people, eventually taking over the processes that humans once completed, at a much faster pace.



Repetitive tasks

Carried out 50-60 times a day



Process List and Storage



Data Entry

Periodic reporting and data analysis



Mass Email Generation

Archiving and extracting



Data Conversion

Conversion of data format and graphics



ERP Transactions

ERP and other back office transactions

PRACTICAL CASE

This example will demonstrate the requisition and purchase process of materials, in RPA this is a process from end to end (requisition to payment) where a sequence of activities are executed by multiple users interacting with multiple systems (mail client, Excel, SAP).

In this case the bots are integrated in almost all stages of the process and through multiple functions.

PROBLEMATIC

The execution of manual tasks cause the process to be slow, subject to deviations, causing bottlenecks in which it is necessary to invest several hours. The extraction of data from different applications or platforms becomes an extensive task in which it is necessary to invest more than one resource in the execution of a simple process.

AREAS INVOLVED

Purchasing, Warehouse and Accounts Payables.

USERS INVOLVED

Requester, Buyer, Vendor and AP analyst.

PROCESS STEPS

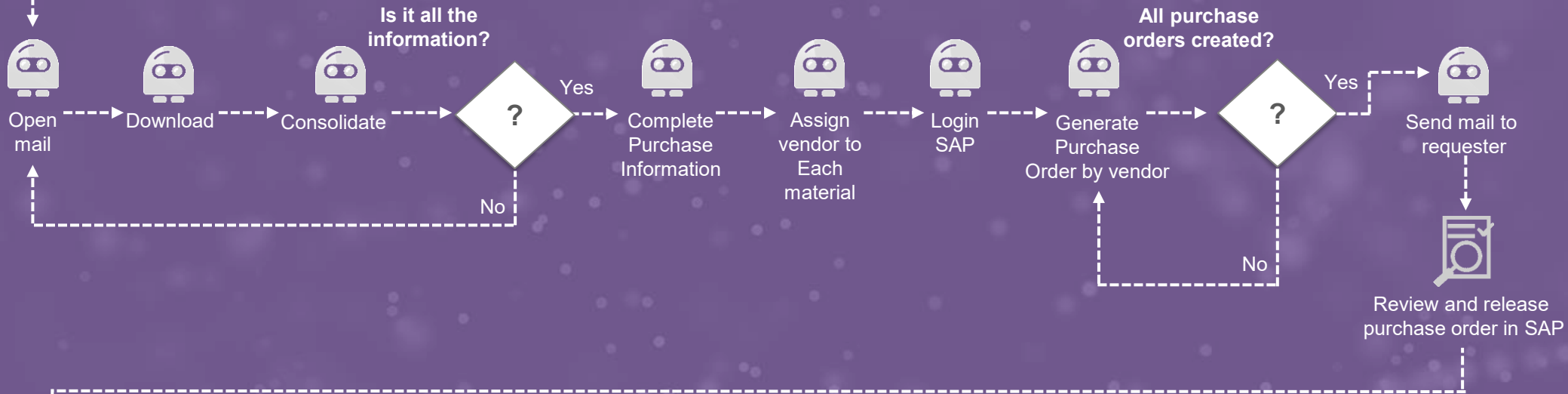
- **Requester: Create in Excel and send the request.**
- **Buyer: Receive and consolidate order requests organized by vendor, create and release purchase order in SAP.**
- **Vendor: Send XML and invoice PDF.**
- **AP Analyst: Extract information from the invoice and register it in SAP.**

MANUAL PROCESS VS PROCESS IN RPAIDE

Requester



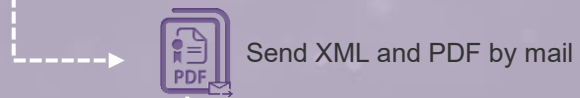
Buyer



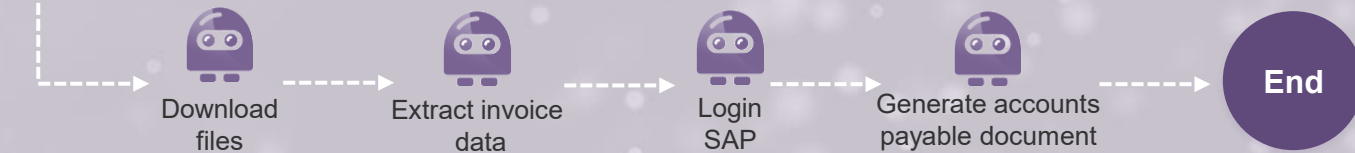
Warehouse



Vendor

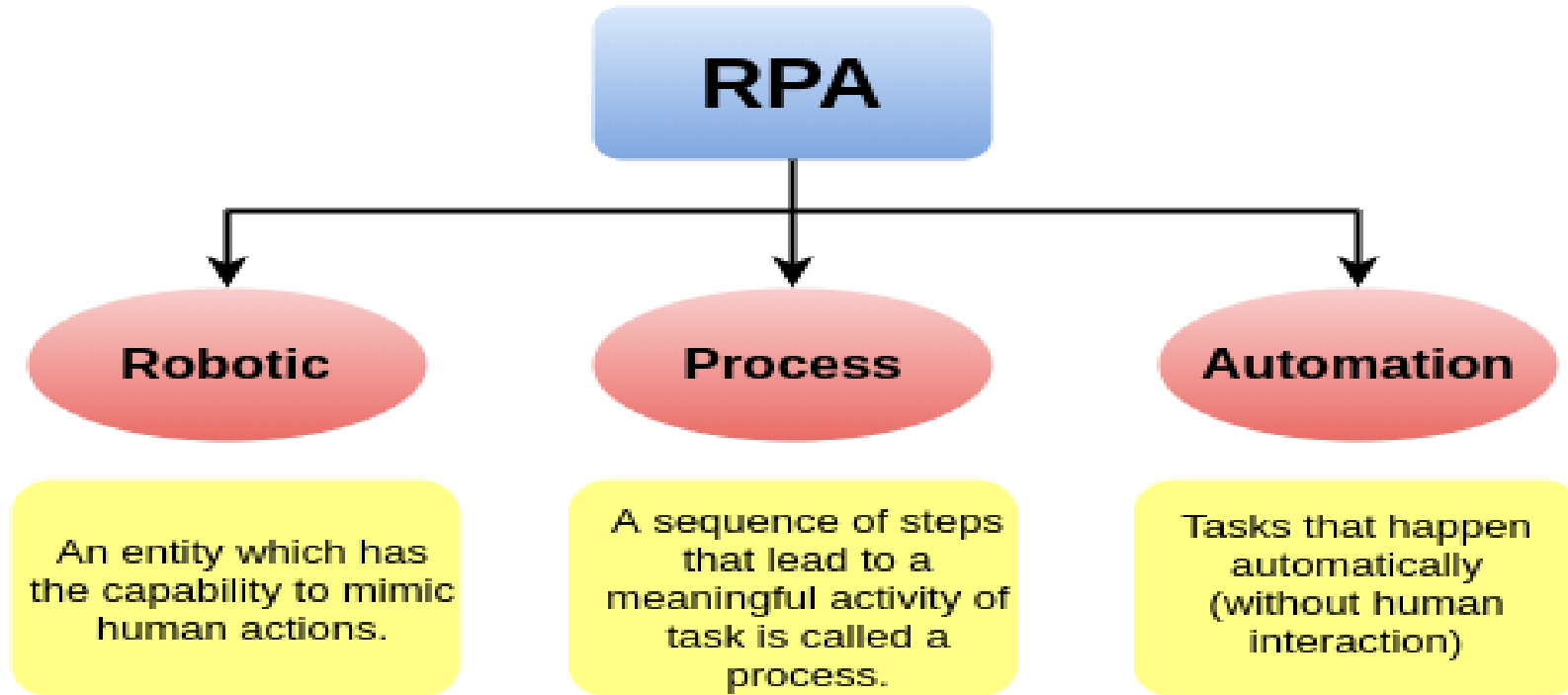


AP Analyst



RPAide
By Emergys

What is RPA

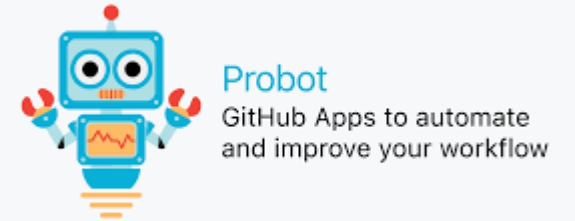


- 'Robot' in 'RPA' is not a physical robot but a virtual system that helps in automating the repetitive manual computing or business process tasks.

RPA technologies

Probots

- These are the bots that follow simple, repeatable rules to process data.



Knowbots

- These are the bots that search user-specified information from the internet and respond to the user.



Chatbots

- These are the bots that act and respond as virtual agents. They reply to customer queries in real-time.

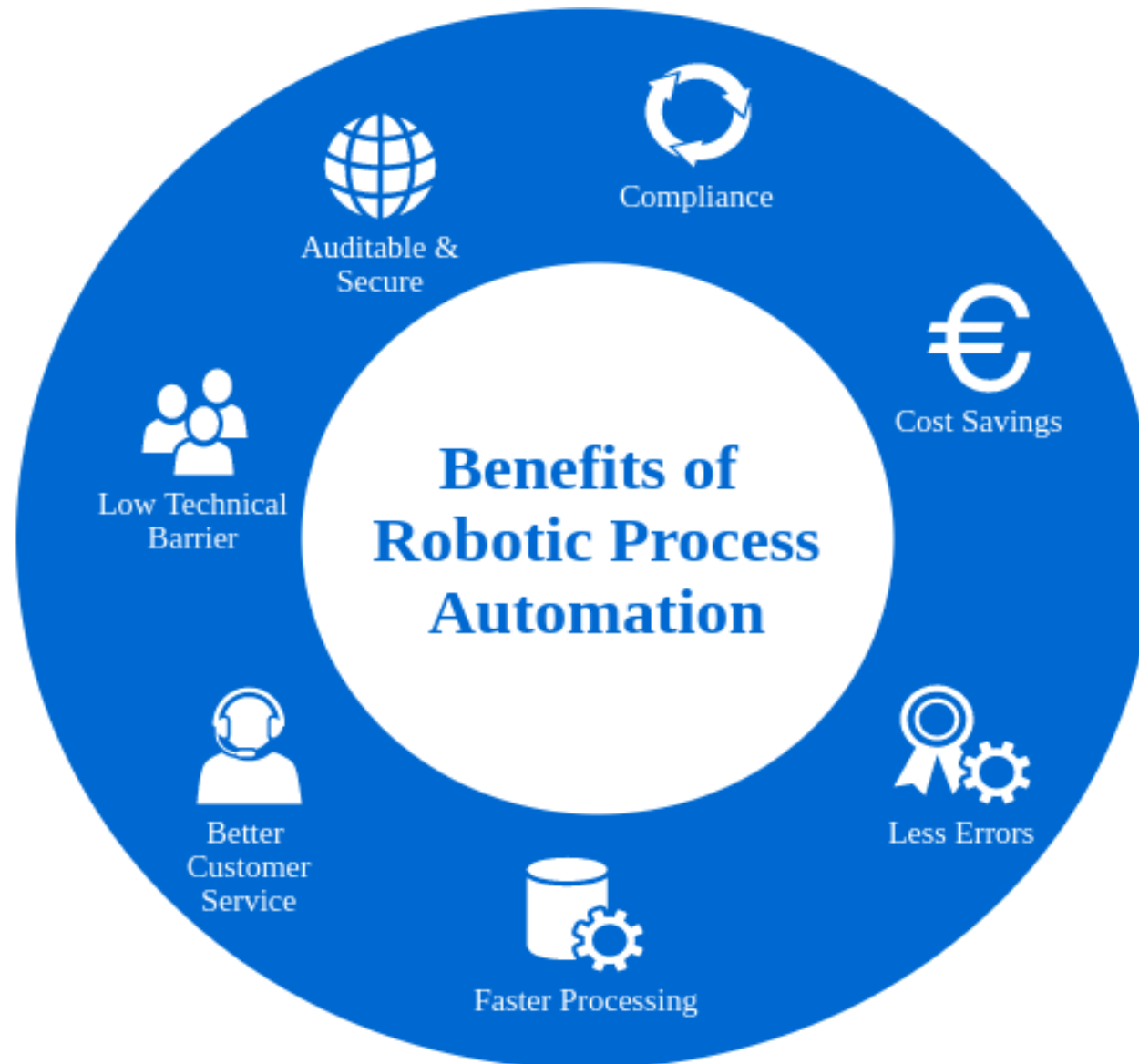


Why RPA

- RPA is not part of an organization's IT infrastructure.
- Instead, it belongs to the top of the whole organization's IT infrastructure.
- In an IT environment, most of the business processes are not smart and intelligent.
- RPA uses bots that interact with web applications, web sites, excel worksheets, and emails to automate the tasks just like a human
- It helps human beings to focus on those tasks which require emotional intelligence, reasoning, judgement, and interactions with the customers, rather than just doing repetitive tasks



Benefits of RPA



History of RPA

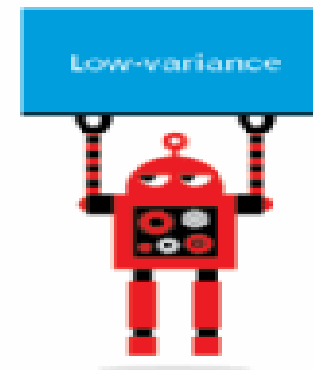
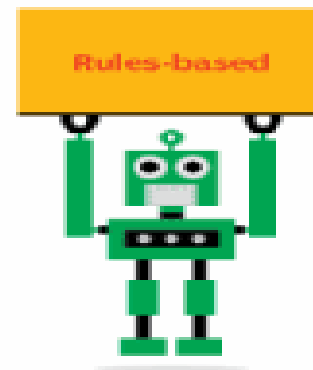
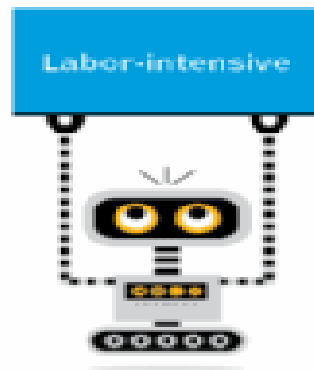
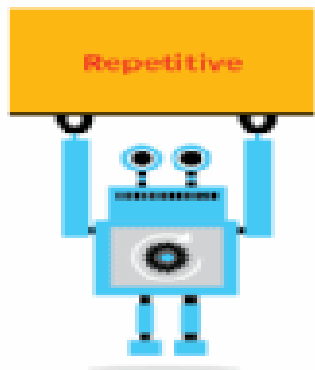
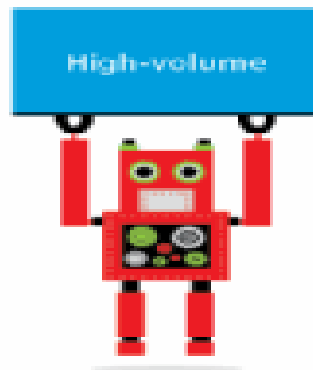
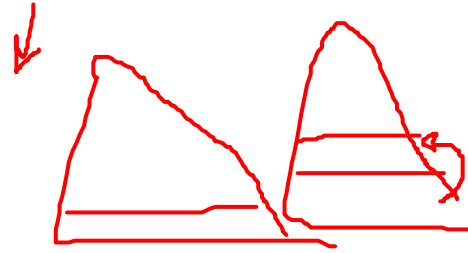
- '**Machine Learning (ML)**' eventually lead to the creation of RPA.
- In 1959, '**Arthur Samuel**' developed Machine Learning. Machine Learning allowed computers to perform several critical tasks, such as translation and text summarization, etc.
- '**Natural Language Processing (NLP)**,' which helped computers to understand and process human language more accurately.
- In 1960, NLP combined '**AI (Artificial Intelligence)**' for establishing the interactions between computers and human languages.
- **Screen Scraping** technology is considered as a significant step towards the creation of RPA. This technology is used to extract data from web, programs, and documents, which is further displayed by another application.
- **Workflow automation** which helps in reducing the human task. These actions must be repetitive and predictable.
- **Artificial intelligence** is the ability of computer machines or robots to perform tasks that typically require human intelligence. AI programming is based on three techniques: **learning**, **reasoning**, and **self-correction**.

Advantages of RPA

- Code-Free
- Non-Disruptive
- User-Friendly
- Rich-Analytical Suite
- Security
- Rule-based Exception Handling
- Hosting and Deployment Options
- Actionable Intelligence
- Debugging

Disadvantages of RPA

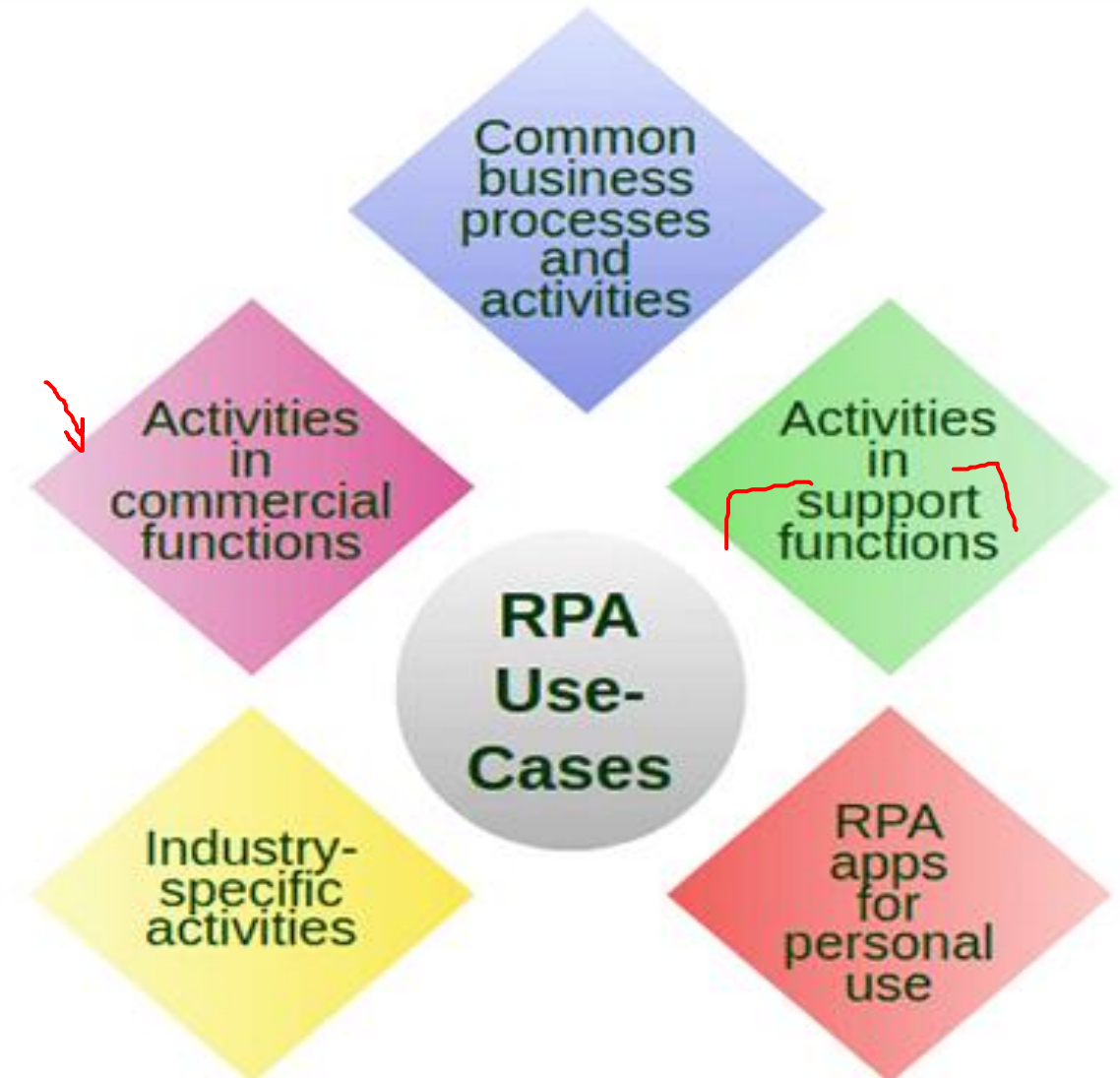
- Potential Job Losses
- Initial Investment Costs
- Hiring Skilled Staff
- Employee Resistance
- Process Selection



RPA Use Cases

1. Common business processes and activities
2. Activities in commercial functions
 - Sales
 - Customer Relationship Management
3. Activities in support functions
 - Tech Support
 - Technology
 - Finance
 - HR
 - Operations
 - Procurement
4. Industry-specific activities
 - Banking
 - Insurance
 - Telecom
 - Retail
5. Robotic Process applications for personal use such as digital assistants

claim

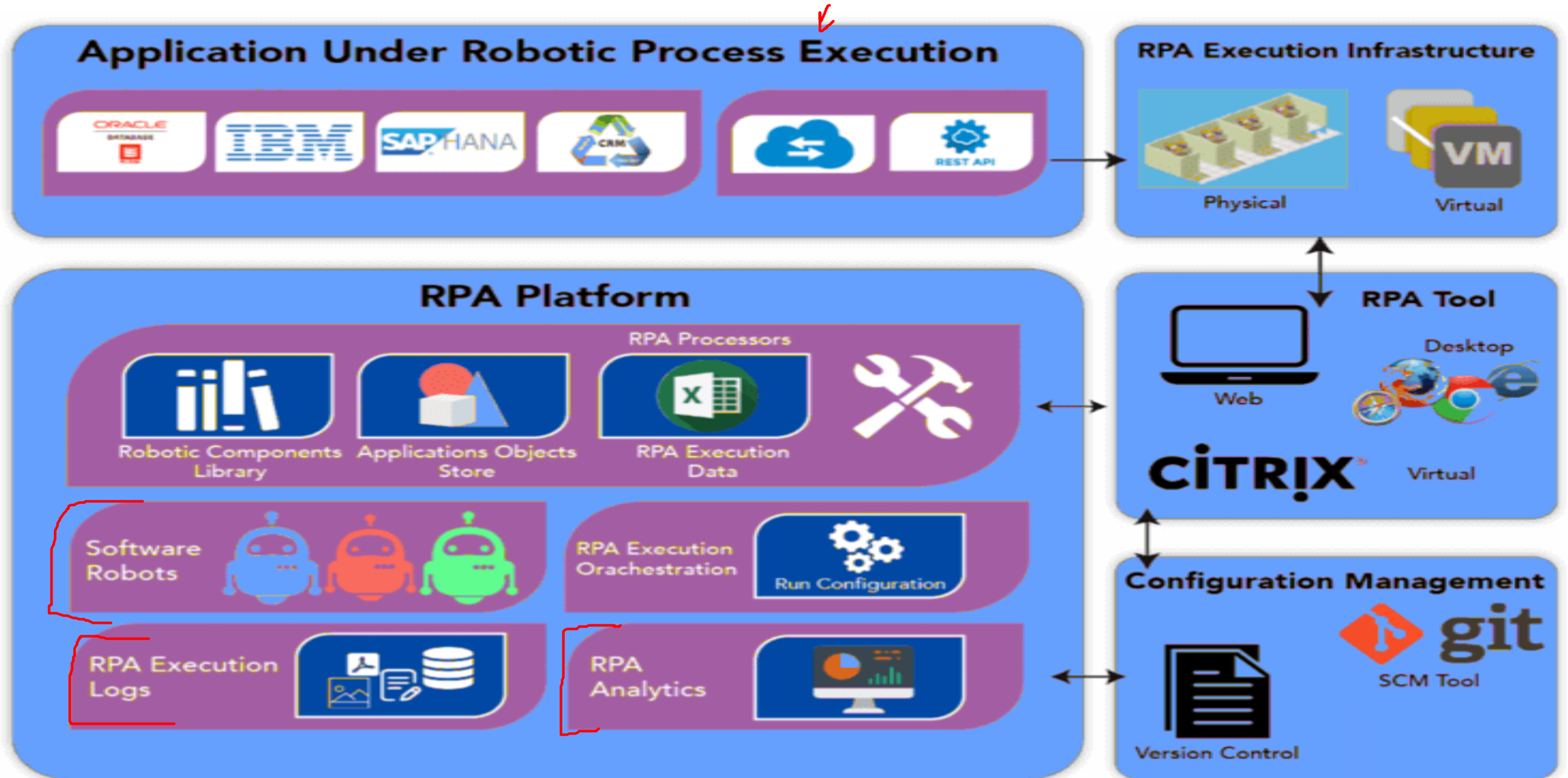


RPA Applications

- Quote-to-Cash
- Procure-to-Pay
- Customer Onboarding
- Employee Onboarding
- Data Migration and Data Entry
- Data Validation
- Extracting Data from PDFs, Scanned Documents and other Formats
- Periodic Report Preparation
- Generating Mass Emails
- Creating and Developing Invoices
- Price Comparison

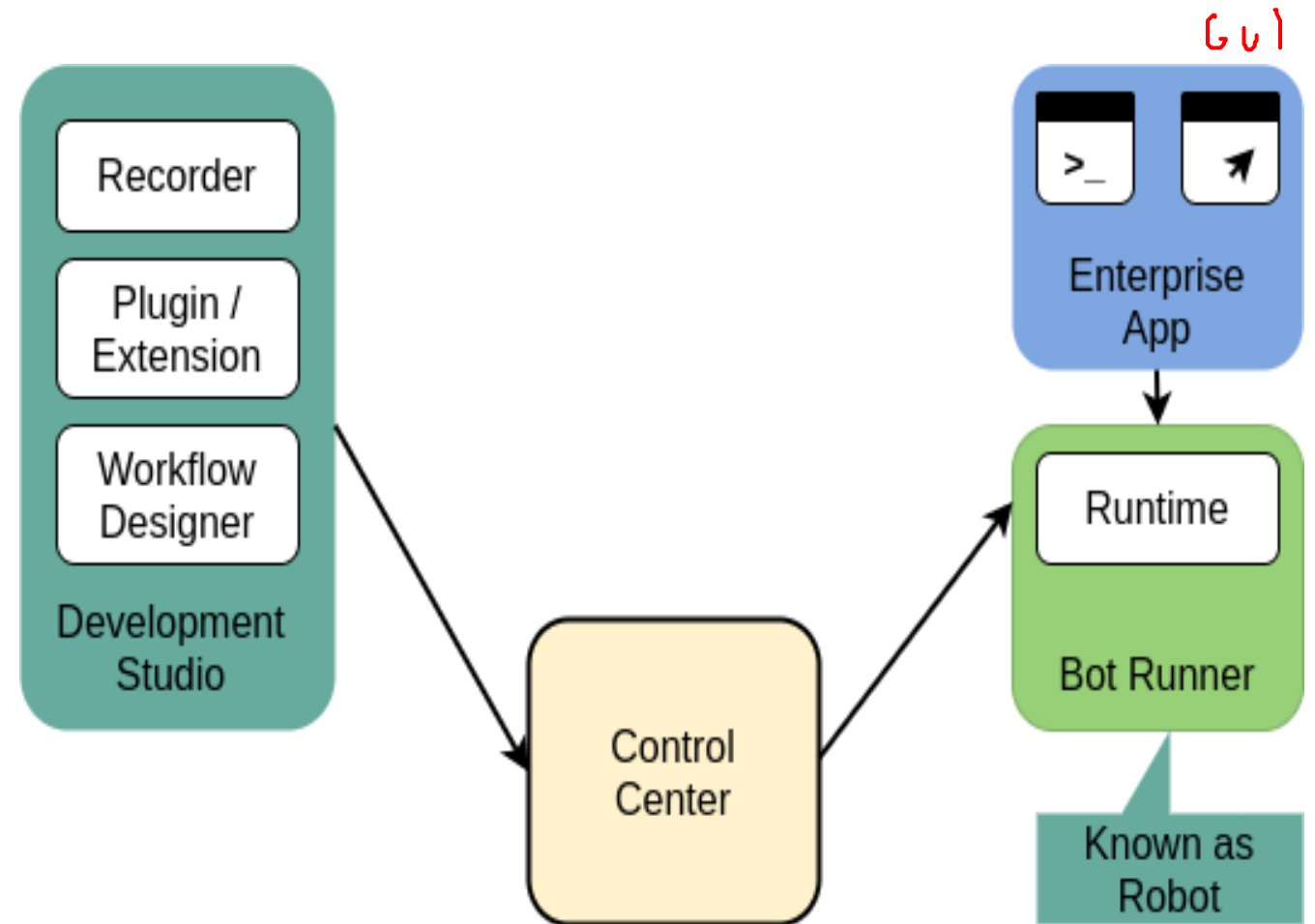
Automating such processes reduces errors and provides fast services

RPA Architecture



Components of RPA

- Recorder
- Development Studio
- Plugin/Extension
- Bot Runner
- Control Center



Components of RPA

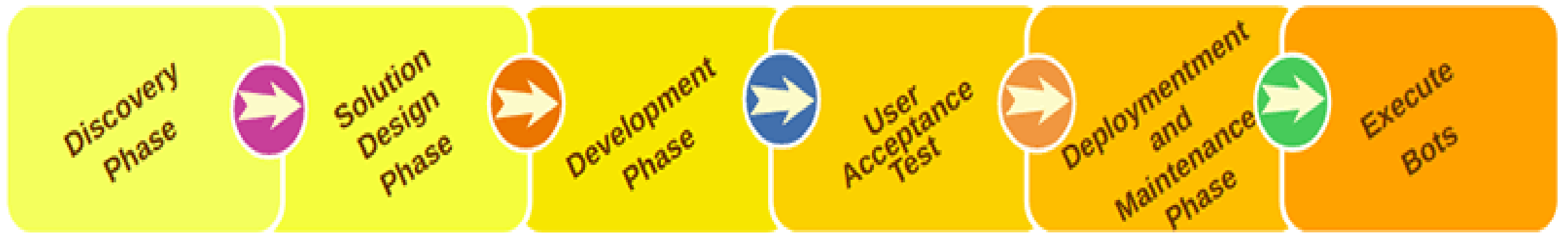
Typical features of RPA Developer Studio

- Dashboard with **GUI (Graphical User Interface)**.
- Different types of Recorders.
- Logging and Exception Handling.
- Integration support with **OCR (Optical Character Reader)**.
- Collection of pre-built, drag-and-drop templates.
- Universal search option to search across all the automation resources such as libraries, activities, projects, etc.

Bot Runner

- are used for executing the developed software bots.
- They are the machines on which bots are run.
- Multiple bots can be assembled parallelly for faster execution.
- The only requirement to run the bots is 'Run License'.
- The bots also report the execution status (i.e., execution logs, pass, or fail, etc.) back to the control center.

RPA Life Cycle



RPA Life Cycle

Discovery Phase

- In this phase, the RPA process architect analyzes the requirements of the client.
- Then it is further decided whether the process can be automated or not. If the process can be automated, then the RPA analyst team might involve the RPA architect team and analyze the complexity of the process.



Solution Design Phase

- Based on requirements, the steps to automate the task, are designed. The RPA technical architect, in collaboration with the process architect, develop a **Process Definition Document** (PDD), which includes information about the whole process. They follow the developmental methodology and develop a strategy to automate specific tasks to reduce manual work as much as they can.
- Once all the requirements are matched, the next step is to decide the budget, number of people, time to be spent on the project, etc. Then, the analyst team creates a flowchart to understand the flow of processes, which helps in choosing the right processes for automation. After the selection of processes, the RPA tool is used to start the development of bots and automate tasks.

Development Phase

- In this phase, the RPA developer creates scripts/bots to automate the tasks with the help of RPA tools. There are several RPA tools available in the market. Automation scripts/bots are generated by following the previously developed PDD. Generally, there is no requirement for coding. But it may change depending on the tasks to be automated.
- After developing the bots, the next step is to test the developed bots.

UAT (User Acceptance Tests)

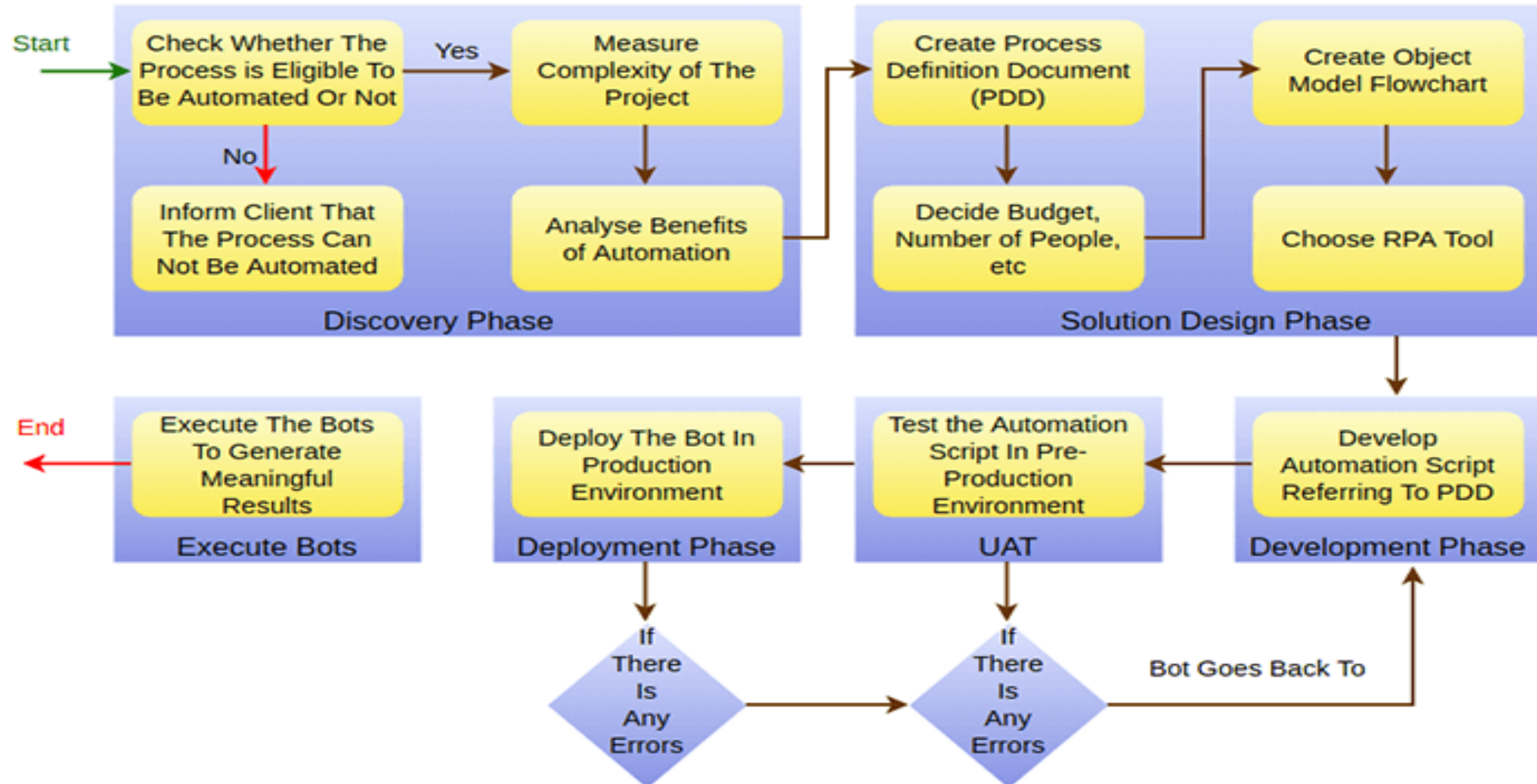
- In this phase, the RPA development team tests the developed bots. These bots are tested in a pre-production environment to examine how the users can use them to automate specific tasks. If the testing phase gets passed successfully, then it is further transferred to the next stage. Besides, if the testing fails, then it is transferred back to the development phase, where RPA developers examine errors found in the testing phase and solve them.
- Once the bots get successfully tested, then they are transferred to the deployment phase of the RPA Lifecycle.

Deployment and Maintenance Phase

- The bots are deployed into the production environment when only they have completed the development and testing phases. After the deployment process, users can use them to automate their tasks. If there is still a problem with the bots, like bots are not automating, then they are further transferred to the RPA development and testing team. The development team will again analyze the bots and resolve the problems.

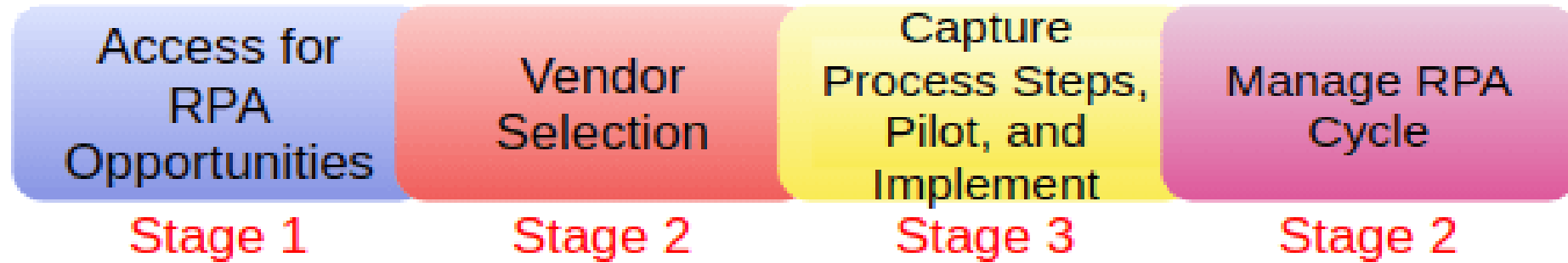
Execute Bots

- This phase includes the execution of the bots after deployment. Bots are also checked to ensure that the implementation is performed as per requirements.



RPA Implementation

RPA Implementation



Keys to Successful Implementation

- organizations should consider the following keys which will help in successful implementation:
 - Select the right project team for internal and external operations.
 - Execute detailed work plans to ensure that no work is dropped.
 - Form quick connections with IT, controls, and procurement.
 - Estimate the impact of changes in people.

How does RPA work?

- In a typical business, the workflow for this process may look like the following sequence:
 - A customer requests for an invoice through mail.
 - The operator checks the mail and opens the relevant billing software.
 - Information is copied from the email into the billing software.
 - The invoice is created using the given information and saved.
 - The original sender is informed that the process is complete.

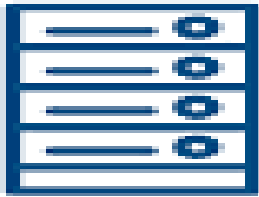


- the RPA tool can automate all of the steps.
- As soon as the customer generates an email request, all these steps will be performed automatically by RPA bots, and there will be no need for human input.
- Preparing and cleansing data in a structured format helps the software bots to easily copy and paste data from one field to another without oversight.
- In case of incomplete, inaccurate, or missing data, these software bots can send the acknowledgment to the original sender and request for the correct data. It prevents all kinds of mistakes that may arise as a result of user error.

How does RPA work with the existing systems?

Back-end RPA Integrations

automation accesses systems and services under the control of a process automation server. It is generally used for unattended automation in which software bots carry out back-office tasks. These tasks may include processing of insurance claims at scale with minimal to no employee intervention.

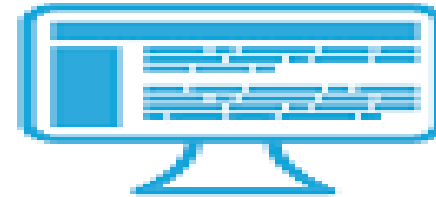


Back End

- Programming and scripting such as Python, Ruby and/or Perl
- Server architecture
- Database administration
- Scalability
- Security
- Data transformation
- Backup

Front-end RPA Integrations

provide several different ways by which automation can connect with desktop applications such as SAP, PeopleSoft, and Salesforce CRM, etc. A front-end automation can read/write data and capture events directly from the user interface of the host application, just like humans.



Front End

- Markup and web languages such as HTML, CSS and Javascript
- Asynchronous requests and Ajax
- Specialized web editing software
- Image editing
- Accessibility
- Cross-browser issues
- Search engine optimization

How does RPA work with the existing systems?

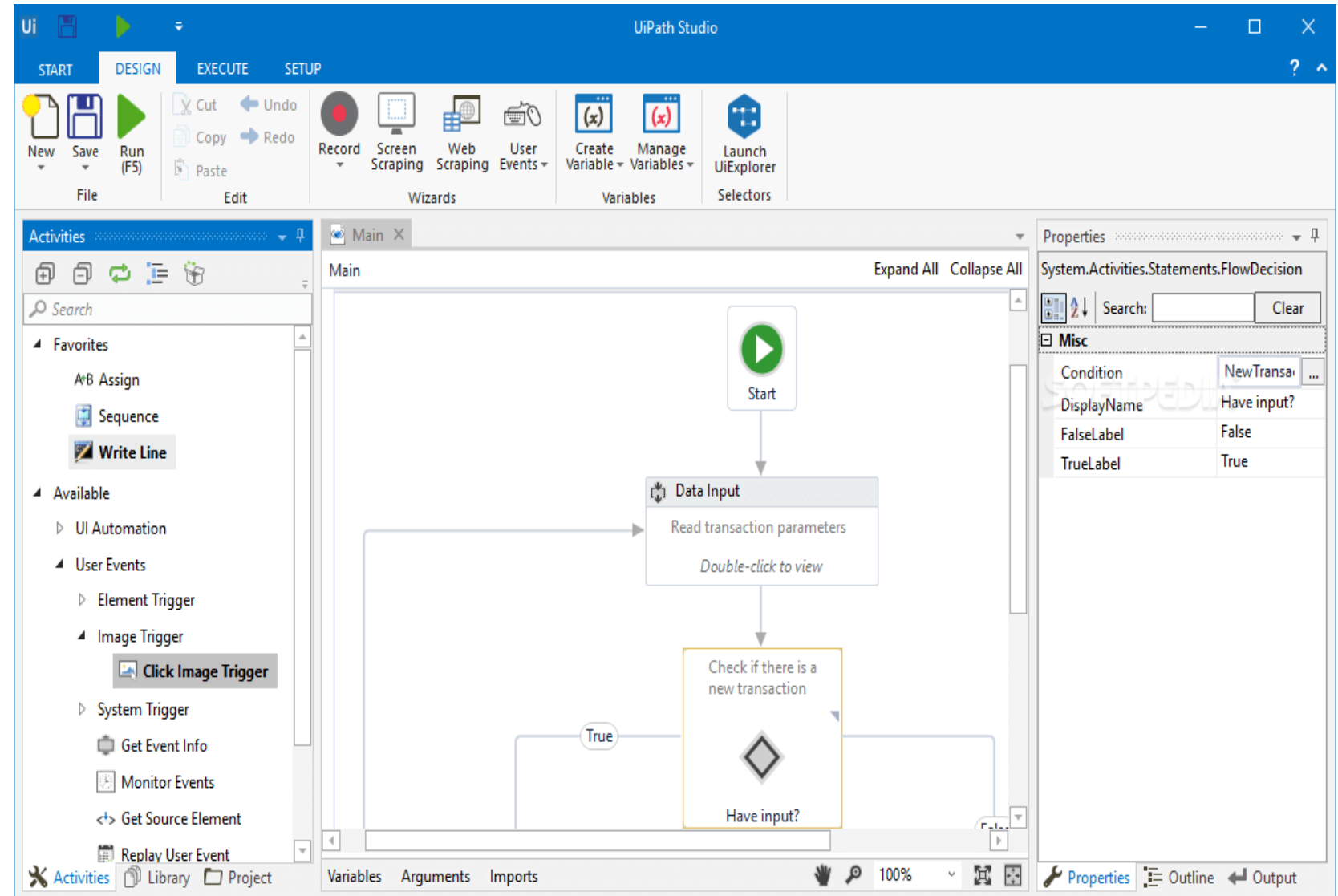
- Working of AI-powered RPA
- leverages several technologies like
- AI (Artificial Intelligence), OCR (Optical Character Recognition), Text Analytics, and Machine Learning, etc. Attended and unattended RPA is also a part of AI-powered RPA technology.
- Machine learning adds abilities to the automation process, which helps in learning, expanding, and continually improving capabilities and certain aspects of its functionality.



RPA Tools

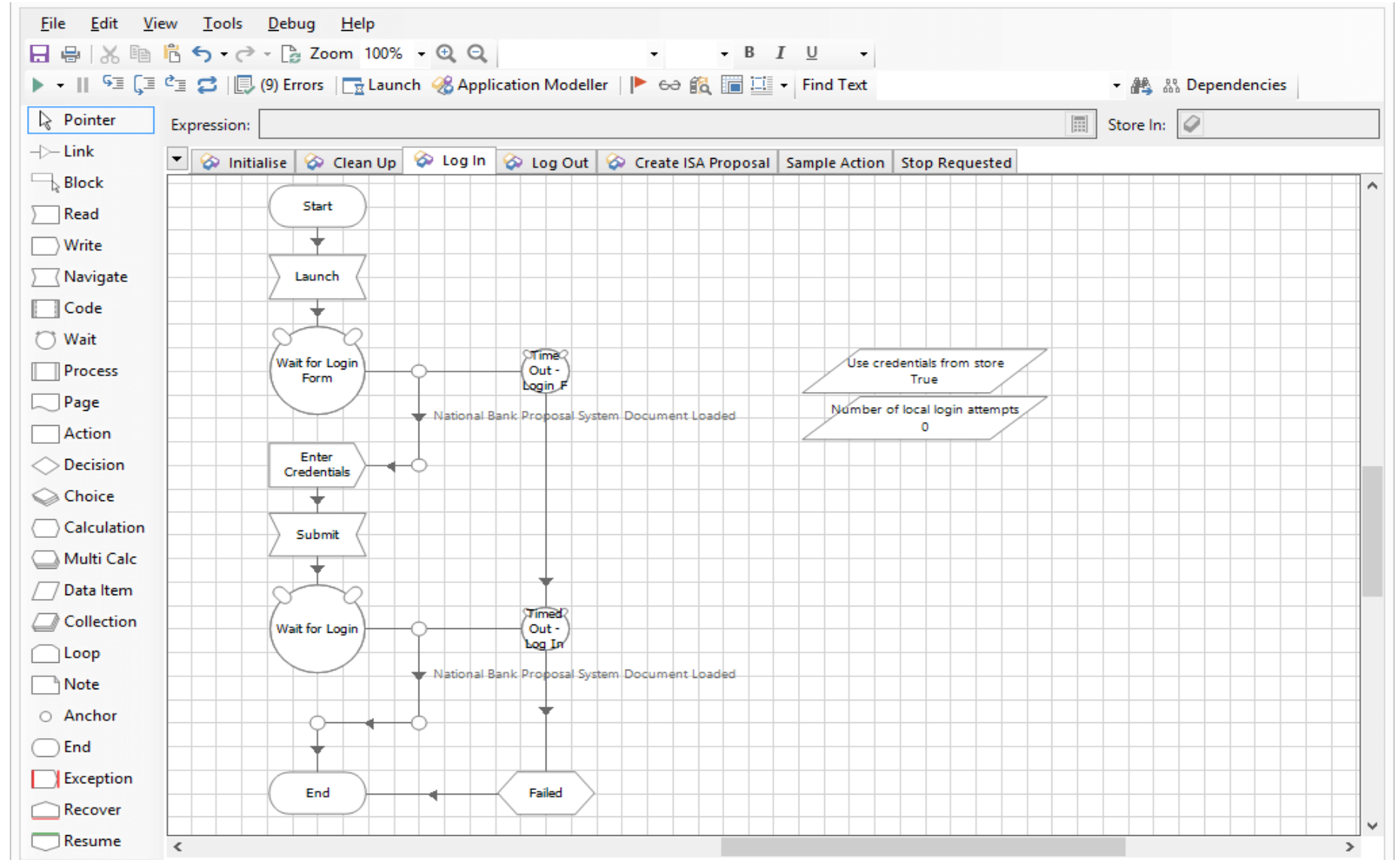
UiPath

UiPath is a highly extensible Robotic Process Automation tool that helps to automate desktop or web applications. It offers global enterprises to design and deploy a robotic workforce for their organizations.



Blue Prism

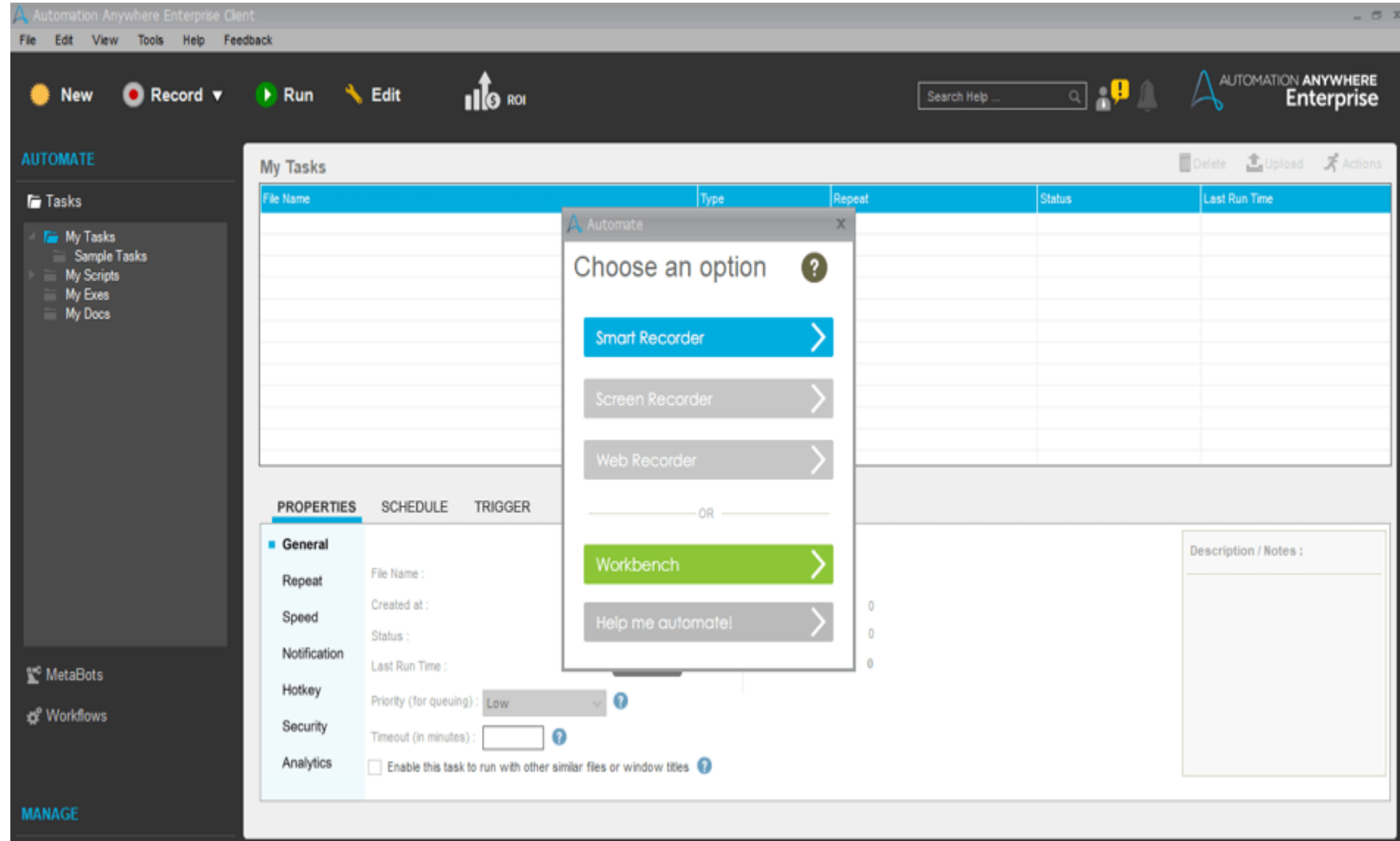
Blue Prism is an RPA Tool that provides a virtual workforce to the organizations. It helps organizations to automate manual, repetitive, and rule-based business processes in an agile and cost-effective manner. It includes drag and drop support to automate the tasks.



Automation Anywhere

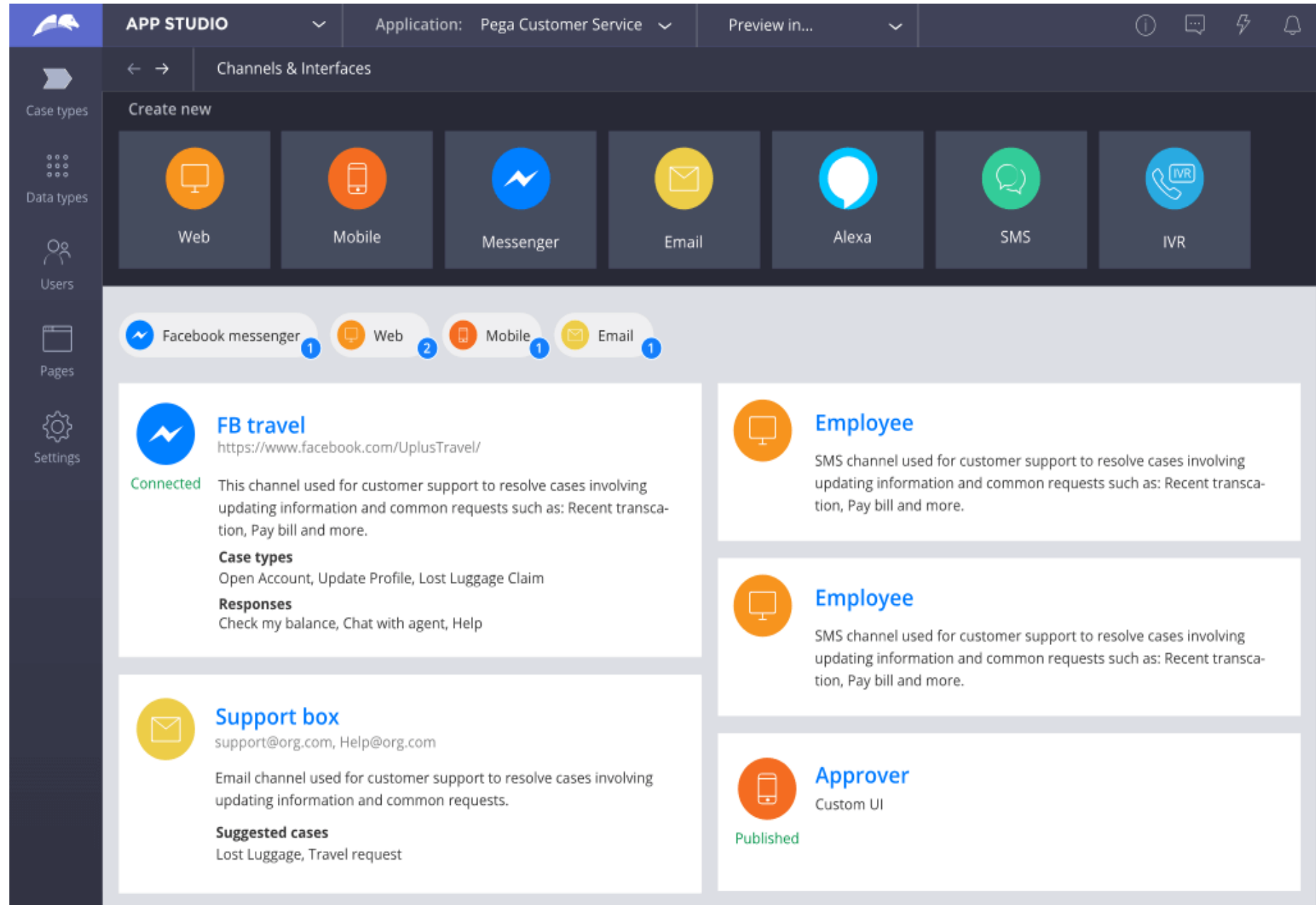
is one of the popular RPA vendors that offers powerful and user-friendly capabilities to automate any complex business process. This tool is a combination of all the core functions. It combines RPA with intellectual elements like language understanding and reading any unstructured data.

It allows automating a broad range of tasks, from basic windows configuration steps to the ultimate networking and remote database processes.



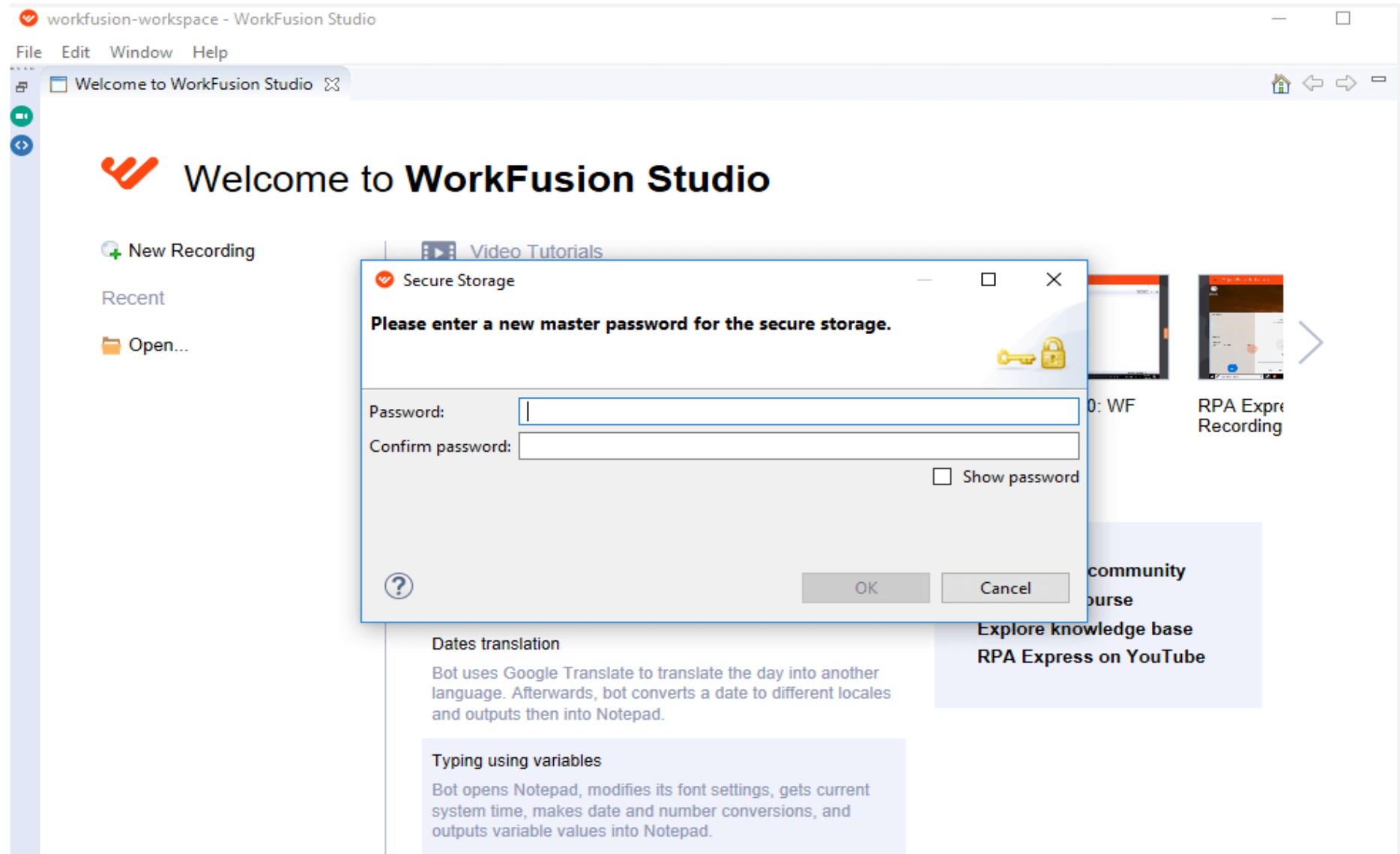
Pega

Pega is an RPA tool that can be used to automate the routine tasks which are repeated in a workflow. It adds the ability to automate tasks using the UI of existing applications. It can be beneficial to speed up manual tasks by automating user actions.



WorkFusion

is a **SaaS (Software-as-a-Service)** cloud computing platform. It automates the monitoring of data feed and helps you to manage crowd-sourced workers from global talent markets.



RPA vs AI

Robotic Process Automation	Artificial Intelligence
RPA is a software robot that can mimic human actions.	AI is the simulation of human intelligence in machines programmed to think like humans and mimic their actions.
RPA robots automate the tasks as per defined rules.	AI is based on 'thinking' and 'learning'.
RPA is a rule-based technology that has no intelligence. It just automates repetitive tasks.	AI includes technologies like ML (Machine Learning) and NLP (Natural Language Processing), which help to do more than just creating rule-based engines to automate repetitive tasks.
RPA can make a huge impact on large businesses as they can process vast amount of data accurately without the need for manual inputting.	AI helps in decision making. It enables the automation of decision making without human involvement.
The main aim of RPA is to automate the repetitive and mundane business processes.	AI replaces human labor. In many companies, physical robots or machines are working in place of human workers.
RPA is easy to implement. Sometimes, an RPA can be complex with large networks of software robots exchanging information between each other, but it still will be a simpler proposition than AI.	AI needs a lot of work to set up and run.
RPA is said to be a process-centric technology because RPA is all about automating repetitive, and rule-based business processes. For example, RPA generates bills or process invoices, and so on.	AI is known as data-driven technology, which is all about providing good quality data. For example, AI helps in reading the bills and invoices and extracting their data to convert it into structured and intelligible information.

RPA vs Testing

Test Automation	Robotic Process Automation
Test automation automates repetitive test cases.	RPA automates repetitive business processes.
Test automation can be applied only to the product and its features.	RPA can be applied to the product as well as other business processes.
Programming or coding knowledge is required to perform Test automation.	RPA provides a drag and drop feature to automate the tasks. Therefore, programming knowledge is not compulsory.
Test automation is implemented across different environments (i.e., QA, Production, Performance, UAT).	RPA only needs a single production environment.
Test automation can be used only by a particular set of users (i.e., Developer and tester).	All the individuals across the team can use RPA.
Test automation can be used with limited software environments. For example, Selenium only works with web applications.	RPA can be used with a wide variety of software environments.
With the help of Test automation, only test cases can be automated.	RPA helps to automate tasks like data entry, loan processing, form-filling, etc.
Test automation helps in reducing the execution time.	RPA helps to minimise human workers.
Test automation works as a virtual assistant.	RPA works as a virtual workforce.
Test automation tools cannot be used for Robotic Process Automation.	RPA tools can be considered as testing tools at a basic level. However, these tools are not as good as test automation tools.
Test automation tools come with a constraint that they require software to run.	RPA can be applied to anything which is in a structured form. Hence, any testing tool exists in the market cannot be used as a RPA tool.

RPA vs Automation

Robotic Process Automation	Traditional Automation
It does not require any modification in the existing systems or infrastructure.	It requires certain customizations in the existing IT infrastructure.
It can automate the repetitive, rule-based tasks. It mimics human actions to complete the tasks.	It does not include the ability to mimic human actions. It only executes the pre-defined programmatic instructions.
A user can start using RPA without knowing any programming. RPA allows automation with easy to use flowchart diagram. Therefore, users do not require to remember language syntax and scripting. They only need to focus on the functionalities given under automation.	Users are required to have the programming skills to use Traditional Automation for automating functionalities. Programming language requirement depends upon the type of automation tool. Users need to remember language syntax and scripting.
RPA provides the easy and quick implementation. It requires less amount of time as RPA software is process-driven.	Traditional Automation can take several months for implementation. Test designing and feasibility studies take a longer time.
RPA allows users to assign work to hundreds or thousands of virtual machines that can perform the allotted tasks without the requirement of physical machines.	On the other hand, Traditional Automation uses different programming techniques to achieve parallel execution or scalability. Physical machines are required to perform parallel execution. Those physical machines should have the capability of providing good processing speed.
RPA can be configured to meet the requirements of a particular user. It can be combined with several applications (e.g., calendar, e-mail, ERP, CRM, etc.) to synchronize information and create automated replies.	When it comes to customization, Traditional Automation is considered as a critical and complex technology compared to the RPA. The integration of different systems with Traditional Automation is a challenge due to the limitations of APIs.
RPA can be a little costly in the initial phase. But it saves a lot of time, money, and effort in the long run.	Traditional Automation is cheaper in the initial phase. However, it costs a lot more in the long run.
RPA is a more efficient option since it can make improvements instantly.	Traditional Automation requires more time, effort, and a

Myths of RPA

- The Robots are Physical / Humanoid
- Robots will replace humans
- Robots are completely accurate
- RPA is expensive
- Robots can automate everything
- Robots are fully automated and can be left unattended 24/7
- To use RPA software, one needs to have basic knowledge of programming



